NIKITA BANWANKAR

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PROFESSIONAL SUMMARY

AI Engineer with hands-on startup experience designing and deploying **Agentic AI applications, Agent Execution Harnesses, and LLM workflows** at Tanglobus. Core expertise in **LangGraph, LangChain, Model Context Protocol (MCP) tool integration, FastAPI, AI Observability (LangSmith/Langfuse), and LLM Evaluation (Evals)**. Proven track record engineering production agents with tool execution loops, state checkpointing, error-recovery middleware, and human-in-the-loop validation.

TECHNICAL SKILLS

Agentic AI & Frameworks: LangGraph, LangChain, DeepAgents, Agent Harness Development, State Management, Checkpointing, Multi-Agent Orchestration, Human-in-the-Loop (HITL)
Protocols & Tool Integration: Model Context Protocol (MCP) Servers & Clients, Structured Tool Calling, Tool Discovery, API Tooling, Context Management
LLM Engineering & Observability: Prompt Engineering, Structured Outputs (Pydantic), Function Calling, RAG, Tracing (LangSmith, Langfuse, OpenTelemetry), LLM-as-a-Judge Evals
Backend, Databases & DevOps: Python (Asyncio, OOP), FastAPI, SQL (PostgreSQL, MySQL), Alembic, Docker, Git, GitHub, REST APIs, JSON Schema

PROFESSIONAL EXPERIENCE

AI Engineer (Agentic AI & LLMs) **Aug 2026 – Present**
Tanglobus (Startup) *Pune, India*

- Architected and developed core **Agentic AI workflows and Agent Execution Harnesses** for the enterprise **APS (Automated Project System) Platform** using Python, FastAPI, LangGraph, and LangChain.
- Engineered custom agent middleware for **tool execution loops, automated retries, error recovery**, dynamic context management, and **human-in-the-loop approval gates** for critical construction and legal milestones.
- Implemented **Model Context Protocol (MCP) tool integrations** and REST API clients, allowing agents to discover tools, execute structured queries across PostgreSQL, and process unstructured engineering documents.
- Integrated end-to-end **AI Observability (LangSmith / Langfuse)** to trace multi-step agent reasoning paths, monitor token consumption, track execution latency, and eliminate tool calling bottlenecks.
- Built automated **LLM Evaluation (Evals) suites** featuring ground-truth benchmark datasets and LLM-as-a-Judge to measure tool selection precision, task completion rates, and eliminate hallucinations.

KEY PROJECTS

Tanglobus-ASP: Multi-Agent Architectural & Workflow Reasoning Engine **Oct 2026**
Python, FastAPI, Google Gemini LLM, LangChain, Pydantic, PostgreSQL, Docker, Pytest

- Engineered an agentic AI reasoning engine for residential architectural projects, featuring role-based context serialization (client vs. staff) and dynamic prompt routing across multiple domain modules.
- Implemented structured tool calling and domain solvers for automated 3D spatial planning, BOQ/cost estimation, construction timeline scheduling, and multi-modal photorealistic AI rendering.
- Designed robust fallback mechanisms and comprehensive test suites (`pytest`) verifying context isolation, tool execution accuracy, and prompt evaluation benchmarks.

LinkedIn Job Copilot AI: Agentic Browser Assistant & DOM Harness **Oct 2026**
Python, JavaScript, Chrome Manifest V3, DOM Tool Calling, LLM API, State Management

- Engineered an autonomous agentic browser assistant that parses DOM context, evaluates candidate skill match in real-time, and executes multi-step web navigation workflows.
- Implemented an agent execution loop with synthetic React DOM event dispatchers and built-in **human-in-the-loop safeguards** that pause execution before final submissions.
- Integrated Google Gemini LLM with structured function calling to dynamically generate tailored application cover notes and answer employer screening questions based on persistent candidate memory.

AI-Powered Multi-Domain Recommendation & Evaluation System **Apr 2026**
Python, Machine Learning, NLP, Scikit-Learn, Pandas, NumPy, Structured Evaluation

- Developed an AI-driven decision engine mapping user attributes to multi-domain career paths using supervised learning and text vectorization, evaluated against custom precision/recall benchmarks.

EDUCATION

Bachelor of Engineering in Artificial Intelligence & Data Science **Aug 2023 – Jun 2026**
Sant Gadge Baba Amravati University (SGBAU), Amravati, India *CGPA: 7.65*

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Application Development Internship Credential | AI & Machine Learning Specialization
Core Competencies: Tool-enabled Agent Design, Agent State & Checkpointing, LLM Tracing & Reliability, Clean Code Architecture